Interior smoothness of weak heat solutions

English theorem, checked Lean statement, and proof outline

Prepared 18 October 2026 • Homogeneous equation, diffusivity one

1. Precise mathematical statement

Write a space–time point as $z = (t, x) \in \mathbb{R} \times \mathbb{R}^n$, and let $P = \partial_t - \Delta_x$. All smoothness below is joint in time and space.

Theorem 1 (Interior regularity on an open cylinder)**.** *Let $n \geq 1$, let $\Omega \subseteq \mathbb{R}^n$ be open, let $T > 0$, and set $Q = (0, T) \times \Omega$. Suppose $u \in L^1_{\mathrm{loc}}(Q)$ and*

$$\int_Q u(t,x)(\partial_t \varphi(t,x) + \Delta_x \varphi(t,x))\, dt\, dx = 0 \qquad \text{for every } \varphi \in C_c^\infty(Q). \tag{1}$$

Then there is a unique $v \in C^\infty(Q)$ such that $u = v$ almost everywhere on Q. Moreover,

$$\partial_t v(t,x) = \Delta_x v(t,x) \qquad \text{at every } (t,x) \in Q.$$

Uniqueness is among smooth representatives of this same almost-everywhere class on Q.

Equation (1) is precisely $Pu = 0$ in distributions. Its plus sign is intentional: integration by parts gives $\langle Pu, \varphi \rangle = -\int u(\partial_t \varphi + \Delta_x \varphi)$. The formalization assumes local integrability, without pointwise differentiability of u.

Stronger proved form. The same conclusion holds for any open $Q \subseteq \mathbb{R}^{n+1}$; a product structure is unnecessary. The cylinder theorem follows from this result.

2. Domain assumptions and boundary scope

Only openness of Ω is needed. It need not be bounded, connected, or have a Lipschitz boundary. Each point of Q has a neighborhood whose closure lies in Q, and the proof works in that neighborhood.

The theorem gives no extension of smoothness to $\partial\Omega$, $t = 0$, or $t = T$, and no uniform derivative estimates as these sets are approached. An irregular boundary may affect such boundary conclusions without affecting interior smoothness. A graph such as $y = \sqrt{x}$ has an unbounded slope in that coordinate system; this alone does not settle whether the surrounding domain is Lipschitz, since boundary charts can be rotated.

The representative qualification is essential: altering u on a null set can destroy pointwise continuity while preserving (1).

3. Proof outline matching the formalization

The following explains the checked proof; the complete formal proof is in the repository modules listed on page 4.

Step 1: Localize and regularize. Fix $z_0 \in Q$. Choose a compact neighborhood $K \subset Q$ and a smooth cutoff χ supported in $\operatorname{int} K$, with $\chi = 1$ near z_0. Extend $u\mathbf{1}_K$ by zero and convolve with normalized smooth bumps whose radii tend to zero. The resulting u_j are smooth, converge to the localized input in L^1 and almost everywhere, and are eventually caloric across the whole compact set $\operatorname{supp}\chi$. The last assertion follows by transferring derivatives to test kernels in the distributional identity.

Step 2: Prove the fundamental-solution identity. Construct the causal Gaussian

$$H(t,x) = \begin{cases} (4\pi t)^{-n/2} \exp\bigl(-|x|^2/(4t)\bigr), & t > 0, \\ 0, & t \le 0. \end{cases}$$

It is locally integrable, smooth away from $(0,0)$, has unit spatial mass at positive times, and approaches the spatial Dirac mass initially. The proved identity, for smooth compactly supported φ, is

$$\int_{\mathbb{R}^{n+1}} H(z)(\partial_t\varphi + \Delta_x\varphi)(z)\,dz = -\varphi(0).$$

Equivalently, $PH = \delta_0$. The proof uses explicit Gaussian derivatives, integration by parts, the initial limit, and Fubini.

Step 3: Remove derivatives of the solution. Apply the fundamental solution to χu_j. The cutoff product rule and spatial integration by parts give, for z near z_0,

$$\begin{aligned} u_j(z) = {} & \int H(z-y)u_j(y)(\partial_t\chi + \Delta_x\chi)(y)\,dy \\ & - 2\sum_{i=1}^{n} \int (\partial_{x_i}H)(z-y)u_j(y)\partial_{x_i}\chi(y)\,dy. \end{aligned}$$

All derivatives on the right act on the cutoff or kernel.

Step 4: Pass to the weak solution. The supports of the differentiated cutoffs avoid z_0. Shrink its neighborhood so that each kernel can be replaced, on all relevant differences $z - y$, by a fixed compactly supported smooth kernel. The right side then defines a smooth function v, since it is a sum of convolutions of integrable functions with smooth compact kernels. Strong L^1 convergence passes the representation to v, and almost-everywhere convergence identifies v with u locally. Thus the smoothness of the limit comes from a fixed convolution representation.

Step 5: Glue and recover the equation. Local smooth representatives agree on overlaps: continuous functions equal almost everywhere on an open set agree everywhere there. Glue them, using a countable subcover to preserve global almost-everywhere equality. The proved classical/distributional equivalence for smooth functions gives $Pv = 0$ pointwise. The same continuity argument gives uniqueness.

4. Exact Lean statement and proof application

The code below is the example extracted from the repository documentation and compiled successfully. Here `Space (n + 1)` is `Fin (n + 1) → ℝ`: coordinate 0 is time and coordinates `i.succ` are spatial. The measure is space–time Lebesgue measure.

```
import PDE.HeatInteriorRegularity

open MeasureTheory Set
open scoped ContDiff BigOperators

namespace PDE.HeatInteriorSpecification

example (n : ℕ) (hn : 0 < n)
    (Ω : Set (Space n)) (hΩ : IsOpen Ω)
    (T : ℝ) (hT : 0 < T)
    (u : Space (n + 1) → ℝ)
    (hu : IsDistributionalHeatSolution (heatCylinder Ω T) u) :
    ∃ v : Space (n + 1) → ℝ,
      ContDiffOn ℝ ∞ v (heatCylinder Ω T) ∧
      u =ᵐ[volume.restrict (heatCylinder Ω T)] v ∧
      (∀ z ∈ heatCylinder Ω T,
        coordDeriv 0 v z = spaceTimeLaplacian v z) ∧
      (∀ w : Space (n + 1) → ℝ,
        ContDiffOn ℝ ∞ w (heatCylinder Ω T) →
        u =ᵐ[volume.restrict (heatCylinder Ω T)] w →
        EqOn w v (heatCylinder Ω T)) :=
  heat_interior_smooth hn hΩ hT hu

end PDE.HeatInteriorSpecification
```

`IsDistributionalHeatSolution Q u` contains exactly local integrability and the test identity (1). The output includes joint smoothness, almost-everywhere equality, the pointwise equation, and uniqueness on the domain. The function v is defined on all of $\mathbb{R}^{n+1}$ in Lean, but its values outside Q are unrestricted by this conclusion.

5. Worked examples

In one spatial dimension, $u(t,x) = x^2 + 2t$ satisfies

$$\partial_t u = 2 = \partial_x^2 u.$$

The checked polynomial example establishes the distributional equation using the proved smooth classical/weak equivalence. It also checks the derivative and sign conventions used throughout the development.

In two spatial dimensions, consider the open cusp

$$\Omega_{\text{cusp}} = \{(x_1, x_2) : 0 < x_1 < 1,\ |x_2| < x_1^2\}.$$

The second checked example proves openness directly from these strict inequalities, then applies the theorem to any distributional heat solution on $(0,1) \times \Omega_{\text{cusp}}$. No boundary regularity hypothesis enters that application. The example does not formalize a separate geometric theorem about non-Lipschitzness of this cusp.

6. Formal implementation and verification

Module family	Role
`HeatDistribution`	Weak equation, a.e. invariance, classical equivalence.
`HeatMollification`, `HeatApproximation`, `HeatApproximationL1`	Smooth caloric approximations and strong L^1 convergence.
`HeatGaussian*`	Gaussian calculus, initial limit, causal smoothness, integrability, and fundamental identity.
`HeatCutoffRepresentation`, `HeatKernelExtension`	Local representation and fixed smooth replacement kernels.
`HeatLocalRegularity`, `HeatInteriorRegularity`	Local smooth representatives, gluing, and the final theorems.

Checked baseline. Commit `4324148`, prepared 18 October 2026. The full library and five example modules passed `bash scripts/check.sh`. All 560 named theorem roots were audited, including 82 added heat roots. Only `propext`, `Classical.choice`, and `Quot.sound` occur; there are no `sorry` proofs or new axioms. This is a Lean kernel build and transitive axiom audit; an independent kernel replay was not performed.

The documented example passed `python3 scripts/check-heat-interior-statement.py`. Lean is pinned to `4.34.0-rc2`; Mathlib is pinned to `85e3a25e006c35636f0e53b0e9296caca2685bc0`. Compilation uses a 3000 MiB Lean budget and a sampled 3500 MiB process-tree RSS guard. No subagents were used.

Solution-class scope. The theorem uses the scalar distributional predicate above. An adapter from the older Bochner `IsIntegratedWeakHeatSolution` API remains separate future work.

Source: GitHub repository.

Branch: `examples/heat-interior-regularity`. Exact proof: `PDE/HeatInteriorRegularity.lean`; examples: `Checks/HeatInteriorExamples.lean`. Repository licensing and attribution are recorded in `LICENSING.md`.